

## CURACAO INTL, Curaçao

WMO: 789880

Lat: 12.189N

Lon: 68.960W

Elev: 9

StdP: 101.22

Time Zone: -4.00 (W04)

Period: 94-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |      | Humidification DP/MCDB and HR |      |      |      |      |      | Coldest Month WS/MCDB |      |     |      | MCWS/PCWD<br>to 99.6% DB |      | WSF   |
|---------------|------------|------|-------------------------------|------|------|------|------|------|-----------------------|------|-----|------|--------------------------|------|-------|
|               |            |      | 99.6%                         |      |      | 99%  |      |      | 0.4%                  |      | 1%  |      |                          |      |       |
|               | 99.6%      | 99%  | DP                            | HR   | MCDB | DP   | HR   | MCDB | WS                    | MCDB | WS  | MCDB | MCWS                     | PCWD |       |
| (a)           | (b)        | (c)  | (d)                           | (e)  | (f)  | (g)  | (h)  | (i)  | (j)                   | (k)  | (l) | (m)  | (n)                      | (o)  | (p)   |
| 1             | 23.4       | 24.0 | 19.8                          | 14.6 | 26.1 | 20.2 | 14.9 | 26.2 | 10.1                  | 27.0 | 9.3 | 27.4 | 3.7                      | 120  | 0.591 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 9             | 5.8                    | 33.6            | 27.6 | 33.0 | 27.6 | 32.4 | 27.2 | 28.9                | 32.4 | 28.4 | 32.0 | 28.0 | 31.7 | 6.8                  | 90   |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 28.0                            | 24.1 | 32.1 | 27.2 | 23.0 | 31.5 | 26.9 | 22.6 | 31.3 | 94.1          | 32.3 | 91.4 | 32.4 | 89.5 | 31.6 | 31.9              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |     | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |      |      |  |
|-------------------|------|-----|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|------|------|--|
| 1%                | 2.5% | 5%  | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |      |      |  |
|                   |      |     | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max  | Min        | Max  |      |  |
| (a)               | (b)  | (c) | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |      |  |
| 9.4               | 8.8  | 8.3 | DB                         | 22.1 | 35.8               | 0.9 | 1.5                                                | 21.5 | 36.9       | 21.0 | 37.7       | 20.5 | 38.6       | 19.9 | 39.6 |  |
|                   |      |     | WB                         | 20.2 | 29.2               | 0.9 | 1.2                                                | 19.5 | 30.1       | 19.0 | 30.8       | 18.4 | 31.5       | 17.8 | 32.4 |  |

## Monthly Climatic Design Conditions

|      |                                                                                  | Annual      | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |      |      |
|------|----------------------------------------------------------------------------------|-------------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|------|
|      |                                                                                  | (d)         | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |      |      |
| (6)  | Temperatures,<br>Degree-Days<br>and<br>Degree-Hours                              | DBAvg       | 28.6  | 27.2  | 27.3  | 27.8  | 28.4  | 29.0  | 29.2  | 29.1  | 29.6  | 29.8  | 29.2  | 28.3  | 27.6 | (6)  |
| (7)  |                                                                                  | DBStd       | 1.24  | 0.90  | 0.76  | 0.87  | 0.75  | 0.75  | 0.71  | 0.73  | 0.79  | 0.98  | 1.14  | 1.10  | 1.01 | (7)  |
| (8)  |                                                                                  | HDD10.0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0    | (8)  |
| (9)  |                                                                                  | HDD18.3     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0    | (9)  |
| (10) |                                                                                  | CDD10.0     | 6773  | 533   | 485   | 551   | 553   | 589   | 576   | 593   | 609   | 594   | 594   | 551   | 546  | (10) |
| (11) |                                                                                  | CDD18.3     | 3731  | 274   | 252   | 292   | 303   | 331   | 326   | 334   | 351   | 344   | 336   | 301   | 287  | (11) |
| (12) |                                                                                  | CDH23.3     | 41716 | 2561  | 2342  | 2876  | 3257  | 3833  | 3864  | 3943  | 4323  | 4323  | 4057  | 3393  | 2944 | (12) |
| (13) |                                                                                  | CDH26.7     | 14716 | 557   | 544   | 782   | 1033  | 1412  | 1503  | 1520  | 1868  | 1954  | 1657  | 1135  | 751  | (13) |
| (14) | Wind                                                                             | WSAvg       | 5.7   | 5.8   | 6.1   | 6.1   | 6.0   | 6.2   | 6.4   | 6.0   | 5.8   | 5.2   | 4.8   | 4.8   | 5.6  | (14) |
| (15) | Precipitation                                                                    | PrecAvg     | 548   | 50    | 26    | 16    | 18    | 21    | 22    | 33    | 39    | 44    | 87    | 112   | 90   | (15) |
| (16) |                                                                                  | PrecMax     | 1174  | 154   | 126   | 113   | 114   | 102   | 103   | 123   | 213   | 194   | 373   | 347   | 385  | (16) |
| (17) |                                                                                  | PrecMin     | 270   | 3     | 0     | 0     | 0     | 0     | 0     | 4     | 1     | 0     | 9     | 2     | 4    | (17) |
| (18) |                                                                                  | PrecStd     | 212   | 33    | 27    | 19    | 23    | 28    | 24    | 22    | 43    | 43    | 77    | 86    | 75   | (18) |
| (19) | Monthly Design<br>Dry Bulb<br>and<br>Mean Coincident<br>Wet Bulb<br>Temperatures | 0.4%        | DB    | 31.2  | 31.2  | 32.0  | 32.2  | 33.1  | 33.2  | 33.2  | 34.2  | 34.8  | 34.0  | 32.8  | 31.7 | (19) |
| (20) |                                                                                  |             | MCWB  | 25.7  | 25.9  | 26.3  | 26.8  | 27.7  | 27.5  | 27.7  | 27.7  | 27.3  | 27.7  | 27.5  | 26.2 | (20) |
| (21) |                                                                                  | 2%          | DB    | 30.4  | 30.7  | 31.2  | 31.9  | 32.2  | 32.8  | 32.5  | 33.2  | 33.7  | 32.9  | 31.9  | 31.0 | (21) |
| (22) |                                                                                  |             | MCWB  | 25.5  | 25.7  | 26.1  | 26.6  | 27.2  | 27.3  | 27.1  | 27.6  | 27.6  | 27.6  | 27.2  | 26.3 | (22) |
| (23) |                                                                                  | 5%          | DB    | 30.0  | 30.1  | 30.8  | 31.2  | 32.0  | 32.1  | 32.0  | 32.7  | 32.9  | 32.1  | 31.1  | 30.2 | (23) |
| (24) |                                                                                  |             | MCWB  | 25.4  | 25.3  | 25.9  | 26.4  | 27.1  | 26.9  | 26.8  | 27.3  | 27.5  | 27.2  | 26.9  | 25.9 | (24) |
| (25) |                                                                                  | 10%         | DB    | 29.2  | 29.5  | 30.1  | 30.9  | 31.2  | 31.5  | 31.4  | 32.0  | 32.2  | 31.8  | 30.8  | 29.8 | (25) |
| (26) |                                                                                  |             | MCWB  | 24.9  | 24.9  | 25.5  | 26.2  | 26.7  | 26.5  | 26.5  | 26.9  | 27.1  | 27.0  | 26.7  | 25.7 | (26) |
| (27) | Monthly Design<br>Wet Bulb<br>and<br>Mean Coincident<br>Dry Bulb<br>Temperatures | 0.4%        | WB    | 27.2  | 27.5  | 28.0  | 28.6  | 28.9  | 28.6  | 28.7  | 29.2  | 29.2  | 29.4  | 28.8  | 27.6 | (27) |
| (28) |                                                                                  |             | MADB  | 30.1  | 30.1  | 30.4  | 31.0  | 32.1  | 32.4  | 32.4  | 33.0  | 33.1  | 32.9  | 31.6  | 30.3 | (28) |
| (29) |                                                                                  | 2%          | WB    | 26.5  | 26.6  | 27.1  | 27.6  | 28.1  | 28.0  | 27.9  | 28.6  | 28.7  | 28.4  | 27.9  | 26.9 | (29) |
| (30) |                                                                                  |             | MADB  | 29.7  | 29.6  | 30.3  | 30.8  | 31.7  | 32.0  | 31.9  | 32.6  | 32.6  | 31.9  | 30.9  | 29.9 | (30) |
| (31) |                                                                                  | 5%          | WB    | 25.8  | 25.9  | 26.4  | 26.8  | 27.5  | 27.5  | 27.3  | 27.9  | 28.0  | 27.7  | 27.2  | 26.5 | (31) |
| (32) |                                                                                  |             | MADB  | 29.1  | 29.2  | 29.9  | 30.4  | 31.2  | 31.4  | 31.3  | 32.0  | 32.1  | 31.2  | 30.3  | 29.4 | (32) |
| (33) |                                                                                  | 10%         | WB    | 25.3  | 25.3  | 25.7  | 26.4  | 26.9  | 26.9  | 26.7  | 27.2  | 27.5  | 27.1  | 26.7  | 26.0 | (33) |
| (34) |                                                                                  |             | MADB  | 28.3  | 28.5  | 29.3  | 29.9  | 30.5  | 30.6  | 30.5  | 31.1  | 31.4  | 30.6  | 29.7  | 28.9 | (34) |
| (35) | Mean Daily<br>Temperature<br>Range                                               | 5% DB       | MDBR  | 5.0   | 5.1   | 5.3   | 5.3   | 5.2   | 5.3   | 5.5   | 5.6   | 5.8   | 5.4   | 4.9   | 4.7  | (35) |
| (36) |                                                                                  |             | MADB  | 5.4   | 5.5   | 5.8   | 5.7   | 5.7   | 5.8   | 5.8   | 6.2   | 6.3   | 5.7   | 5.3   | 5.0  | (36) |
| (37) |                                                                                  |             | MCWBR | 2.8   | 2.9   | 3.1   | 2.8   | 2.9   | 2.8   | 2.9   | 3.0   | 3.1   | 3.1   | 2.9   | 2.7  | (37) |
| (38) |                                                                                  | 5% WB       | MADB  | 5.2   | 5.4   | 5.7   | 5.6   | 5.6   | 5.7   | 5.8   | 6.0   | 6.1   | 5.7   | 5.2   | 4.9  | (38) |
| (39) |                                                                                  |             | MCWBR | 3.2   | 3.4   | 3.6   | 3.2   | 3.3   | 3.2   | 3.3   | 3.6   | 3.6   | 3.6   | 3.1   | 3.0  | (39) |
| (40) | Clear-Sky<br>Solar<br>Irradiance                                                 | taub        | 0.388 | 0.411 | 0.423 | 0.488 | 0.476 | 0.503 | 0.484 | 0.442 | 0.428 | 0.404 | 0.394 | 0.396 | (40) |      |
| (41) |                                                                                  | taud        | 2.552 | 2.463 | 2.439 | 2.233 | 2.304 | 2.210 | 2.278 | 2.439 | 2.467 | 2.565 | 2.583 | 2.556 | (41) |      |
| (42) |                                                                                  | Ebn at Noon | 916   | 905   | 897   | 831   | 827   | 795   | 813   | 859   | 879   | 900   | 904   | 899   | (42) |      |
| (43) |                                                                                  | Edh at Noon | 101   | 114   | 119   | 145   | 133   | 144   | 135   | 117   | 114   | 102   | 97    | 99    | (43) |      |
| (44) | All-Sky Solar<br>Radiation                                                       | RadAvg      | 5.22  | 5.81  | 6.12  | 6.15  | 6.07  | 6.19  | 6.30  | 6.45  | 6.18  | 5.47  | 4.91  | 4.86  | (44) |      |
| (45) |                                                                                  | RadStd      | 0.21  | 0.19  | 0.31  | 0.22  | 0.24  | 0.23  | 0.17  | 0.24  | 0.26  | 0.30  | 0.48  | 0.25  | (45) |      |

## Historical Trends

|                             | DBAvg | Heating |       |        | Cooling |       |       | Degree-Days |         |         |         |
|-----------------------------|-------|---------|-------|--------|---------|-------|-------|-------------|---------|---------|---------|
|                             |       | 99% DB  |       | 99% DP | 1% DB   | 1% WB | 1% DP | HDD10.0     | HDD18.3 | CDD10.0 | CDD18.3 |
|                             |       | (d)     | (e)   | (f)    | (g)     | (h)   | (i)   | (j)         | (k)     | (l)     | (m)     |
| (46) Station Only           | N/A   | -0.34   | -0.58 | -0.53  | -0.77   | -0.70 | N/A   | N/A         | N/A     | N/A     | N/A     |
| (47) Regional (2 neighbors) | N/A   | N/A     | -0.33 | -0.24  | -0.48   | -0.48 | N/A   | N/A         | N/A     | N/A     | N/A     |

Nomenclature: See separate page